Rational Number Project

Fraction Operations and Initial Decimal Ideas **Lesson 11: Overview**	**Materials**
Students estimate sums and differences using mental images of the 10 x 10 grid. Students develop strategies for adding and subtracting decimals using decimal +- board. Students find exact answers to decimal addition and subtraction using mental math.	• Student Pages A -D • Orange and yellow crayons • Teacher Pages 1 and 2

Teaching Actions

Warm up

Imagine adding $\frac{1}{2}+\frac{1}{4}$ using fraction circles. Describe how you would solve this.

Imagine $\frac{1}{2}$ as a decimal, how much of the 10 x 10 grid is shaded? Describe this amount if you used only orange. Imagine $\frac{1}{4}$ as a decimal, How much of the 10 x 10 grid is shaded? How many orange bars and yellow squares would be shaded to show $\frac{1}{4}$?

Large Group Introduction

1. Show the top half of the Addition/Subtract board (Teacher Page 1. Explain: I want you to <u>imagine</u> the decimal, 0.37 on the left grid. What does it look like? Imagine 4 –tenths on the right grid. If you combined the two amounts would you have more or less than $\frac{1}{2}$ of a grid shaded? (Show the bottom of the board with 5 orange bars shaded). Would you have more or less than a whole grid shaded? Explain your thinking. [At this point you are not showing the decimals on the board. This is all mental math at this point].

2. Repeat for the 3 problems shown on Teacher Page 2. Emphasize using $\frac{1}{2}$ as a benchmark.
    - .56 take off .009

Comments

This is a 2 -day lesson.

The initial model for decimals in this module is the 10 x 10 grid.

In this lesson you are developing estimation strategies based on students' mental images related to the 10 x 10 grid.

Another indicator of decimal understanding is students' ability to use their understanding of the relative size of decimals to guide their estimation with operations with decimals.

Students use their mental images for decimals to guide their estimation. Consider how these two students demonstrated this ability:
"Picture 57-hundredths on the Decimal +/− Board. If you took away 9-thousandths would the amount left shaded be more than ½ or less than ½? Explain without finding the exact answer".
Student responded:
S: More than ½.
I: And how do you know that?
S: Because these 9 thousandths would only take up one of the hundredths because you just split one of those into 10 and it would be 9 and so it would be 5 hun...yeah, 561. thousandths.

Teaching Actions

- .16 add .7
- .758; take off .6

Small Group/Partner Work

3. Have students use the Decimal +- board on Student Page A to model each estimation problem and find the exact answer. Let students find their own way to use the board to add and subtract decimals. See the teacher notes for this lesson for examples of how students might do these types of problems.

4. It is less obvious ho to use the board to subtract. Step back and see how students use the board for subtraction. (See Teacher Notes for this lesson to see how students use the decimal +- board).

Wrap Up

5. Have students share their strategies for solving each addition and subtraction problem. Be purposeful in selecting students so correct strategies are modeled. Look for different ways in which students used the boards to add and subtract.

6. If you see common errors make those public as well.

Comments

Solving the same task another student relied on his skill in decomposing decimals, his mental images of the 10 x 10 grid, and then reconfiguring (composing) the decimal parts into an estimate of the final amount.

S: Bigger. Because 9-thousandths is like, like 9 itty-bitty yellows and that's only taking away like 1 (hundredth) away from that (*57 hundredths*). Probably not even one but then you would, it would still be 56-hundredths and something.

From our first teaching experiment we found this model to be quite effective to build mental representations for decimals. On the decimal assessment, the percent correct on the 6 order items ranged from 84% to 100%.

But the assessment also showed that students did not do as well on decimal addition and subtraction items. Whole number thinking was still prevalent among many students. With the first group of 6th graders the percentage correct for 6 decimal operation problems was from 35% to 59%.

We considered reasons for this and decided that the 10 x 10 grid model needed to be adapted to more explicitly show the action of addition and subtraction. We used in the second teaching experiment a "decimal +- board". (see Student page A). We found that in the second teaching experiment students resonated to this model; they used it to solve the problems; they were able to describe verbally how to add and subtract decimals using this board. On the decimal assessment students' performance increased. With the second group of 6th graders the percentage correct for decimal

Teaching Actions

Day 2

Large Group Instruction

7. Return to each problem from the previous lesson. Ask students how they might record what they did with the board using written symbols. Compare the exact answer using symbols with the answer obtained using the board.

8. Record like this:
   .37
   .4

   Students should verbalize that putting tenths below tenths in the example above matches adding 3 tenths to 4 tenths on their addition/subtraction board (or 3 orange strips to 4 orange strips). Please do not tell students the rule: add zero and add as whole numbers!! Students will construct meaningful strategies and do not need a rote rule.

9. Explain to students that another student said that the answer to: .34 + .5 = .39. Ask: Is that reasonable? What is the error?

Small Group/Partner Work

10. Ask students to complete Student Pages B and C. Remind students to estimate before they find the exact answer. Encourage students to represent each decimal based on its place value components (ex: .75 would be shown as 7 orange bars and 5 yellow squares).

11. Students who have been successful with adding and subtracting decimals are comfortable working with decimals in their place value components. They are able to decompose the decimal to operate on them and then compose the parts to name the decimal using accurate decimal language. This is another indicator of decimal understanding: **Using a model and their ability to compose and decompose decimals to interpret addition and subtraction operations and build meaning for work with symbols.**

Comments

operations ranged from 93% to 100% on 4 operations problems using the decimal board.

We believe this board was helpful because students were able to clearly show each number in the problem before they added or subtracted them.

The mental images students have of the addition/subtraction grid help students overcome whole number thinking. Below find two 3 examples of students' thinking on subtraction problems students were asked to solve mentally:

.55 - .3 So this would be 25 hundredths. I just know 5 tenths minus 3 tenths would be 2 tenths and this extra 5 hundredths.

2.3 - .05 I don't think I have done a problem like this. (Encouraged to imagine the grid) 2 and 3 tenths would be 2 things colored in and 3 tenths and then 5…. Now I got it. It would be 2 and 25 hundredths. I was imagining this being 2 and 30 hundredths and 30 hundredths minus 5 hundredths would be 25 hundredths.

2.3 - .05 I imagine this one with the grid so 2 full grids and then there's 3 tenths so you minus 5 of the hundredths, so then you cross out 5 of the hundredths which leaves you with 25. 2 and 25-hundredths.

Teaching Actions

Wrap Up

5. End class with mental math problems. Ask students to imagine the addition/subtraction board to mentally determine the exact answer to each problem. Ask students to explain their thinking.

    - .37 + .6
    - 1.02 + .08
    - .46 + .15
    - .50 - .05
    - .82 -.09

Comments

Encourage students to be precise in their language. While this student was able to estimate and find the exact answer mentally, her language was imprecise and reflected whole number thinking.

> 28 hundredths + 6-hundredths. That would be less than 1-half because 8 plus 6 equals 14 and there's no tenths. It is going to make it 34.

This next example shows a student thinking in terms of decimals. The problem was 2.3 - .05

> I imagine this one with the grid so 2 full grids and there there's 3 tenths so then you minus 5 of the hundredths, so then you cross out 5 of the hundredths which leaves you with 25. 2 and 25 hundredths.

Below find examples of a sixth grader's thinking on addition and subtraction tasks. This student was identified by her classroom teacher as one who struggles in math class. Notice how the use of the decimal +- board helped her develop understanding for decimal addition and subtraction. Student still reverted to "Point" language that should be addressed.

Imagine the decimal board. Use that image to solve this problem: .26 + .5

> 26- hundredths. That would be 2 of the tenths columns and 6 of the small hundredths pieces, the small boxes. You add to it a half of the other grid. Put them together. Point 75.

.55-.3

> That's half a grid and 5 more

Teaching Actions	Comments
	squares. Then take 3 of the columns, tenths columns, minus that. So that would be point 25. There's half of the grid and 5 pieces and I take away 3 from the half and just keep the 5. 2.3 -.05 Still going to be 2 so don't bother with that. Point 3 is 3 tenths columns minus point zero five is 5 little squares. So 3 tenths columns minus 5 little squares (pauses to think) is point 25.

Translations:

- Symbolic to verbal
- Symbolic to picture to verbal to symbolic
- Symbolic to picture

Additional Teacher Notes
Lesson 11

The examples below show students' use of the Addition/Subtraction boards. These examples came from student interviews conducted after completing the decimal lessons.

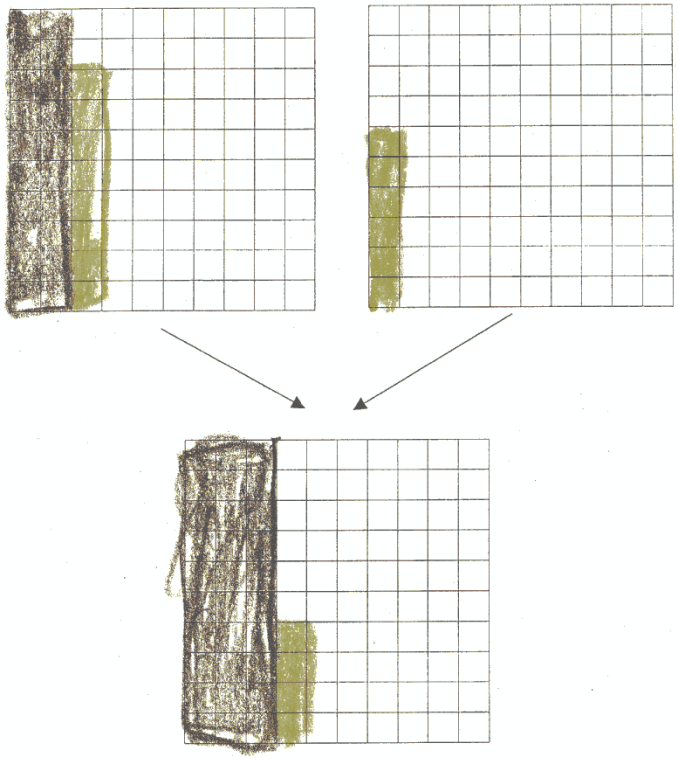
Decimal Addition

.28 + .06

To solve using the board the student first showed .28 on the left grid as 2 orange strips and 8 yellow squares. He showed .06 as 6 yellow squares on the right grid.

He then said, "I take 2 of those to make whole, 3 tenths. Then if you minus 2 from these (6 yellow squares) you have 4 left.

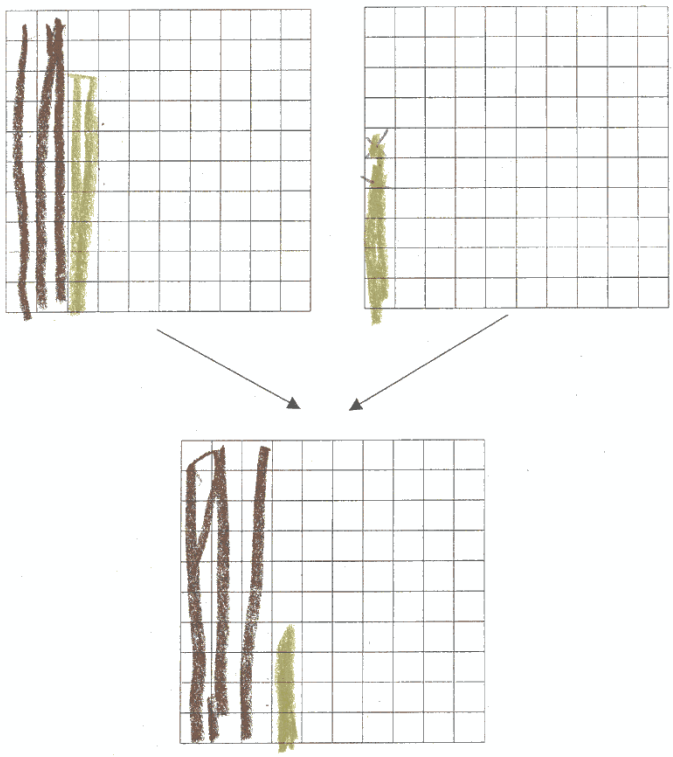
He shows the final answer on the bottom grid as 3 orange strips and 4 yellows.



Notice that the student mentally operated on the display of the two numbers in the problem to determine how to represent the answer on the bottom grid. In the next example, the student solving the same addition problem physically acted out the addition on the grid and then showed the result on the bottom grid.

To solve using the board the student first showed .28 on the left grid as 2 orange strips and 8 yellow squares. He showed .06 as 6 yellow squares on the right grid.

The student then crosses off 2 of the yellow squares from 6-hundredths. She then shades in 3 orange strips and 4 yellows on the bottom grid. "This is 28 so I shade in 2 oranges and 8 yellows. Then there's only 6-hundredths so shade in 6 then take away 2. And add them to that. I am going to make tens so I can make 3. And there is still 4 left over.



Subtraction:

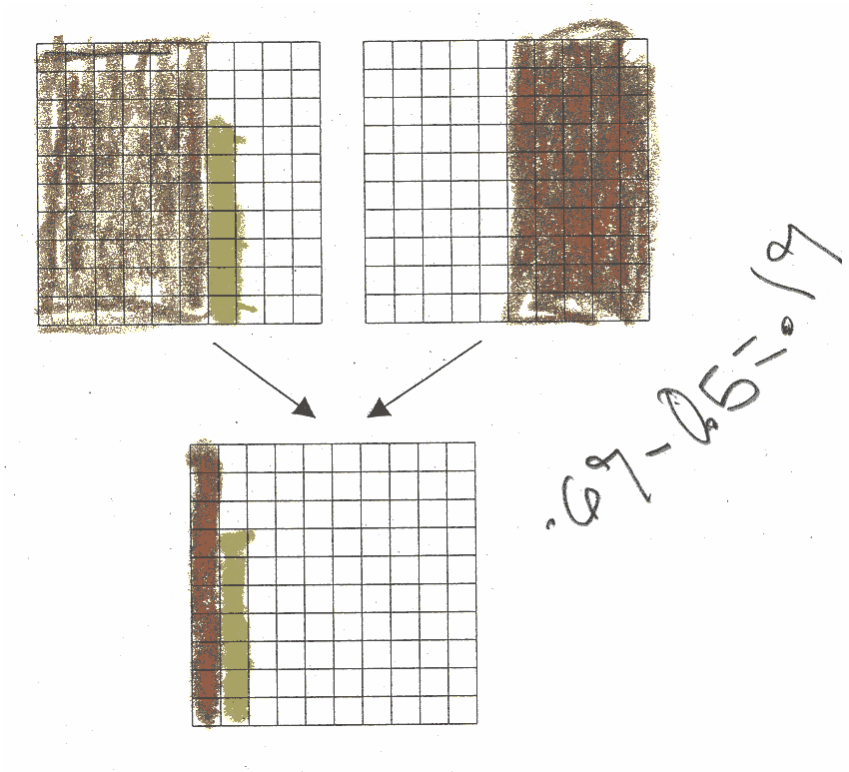
In this example the student acts out the subtraction as take-away marking off the needed number of tenths on the grid on the left side.

.67 -.5

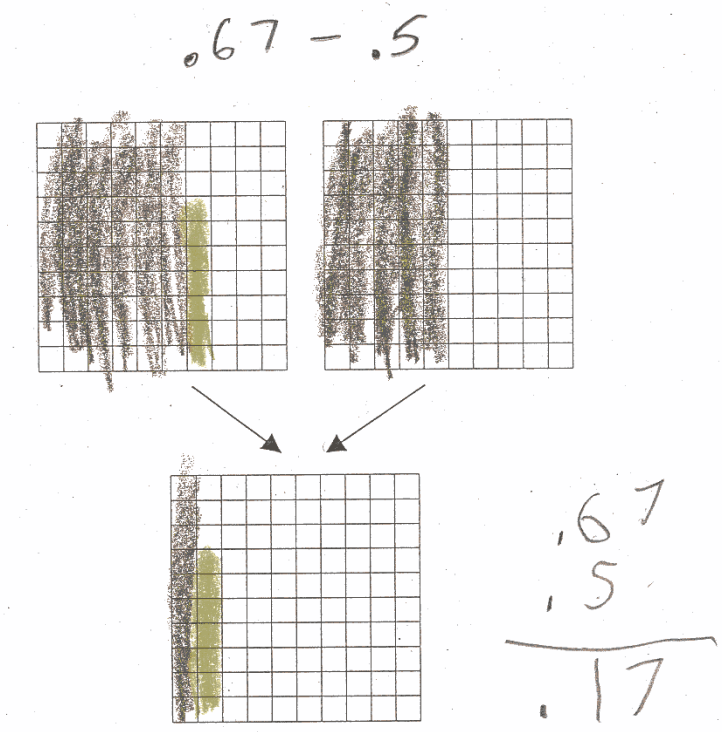
To solve using the board the student first showed .67 on the left grid as 6 orange strips and 7 yellow squares. He showed .5 as 5 orange strips on the right grid.

He takes away 5 tenths by crossing out 5 orange strips on the left grid.
He then shades the answer on the bottom grid as 1 orange strip and 7 yellow squares.

"And I take the 5 oranges away from this and I cross them out over here and then I know when I come down over here I have 1 tenth and 7 hundredths".



Another student showed the two amounts on the board but mentally subtracted 5-tenths and then showed the final answer on the bottom grid. "OK so you would subtract 6 bars from 5 bars which would give you one left and there's 7 yellows, there's 7 squares and no squares so you would just put these squares over there and there you go." (When asked what his final answer was he stated: 17 hundredths."



Notice that this student rewrote the problem vertically to show his work with the grid. He lined up the 5 tenths under the 6 tenths without having to rewrite .5 as .50.

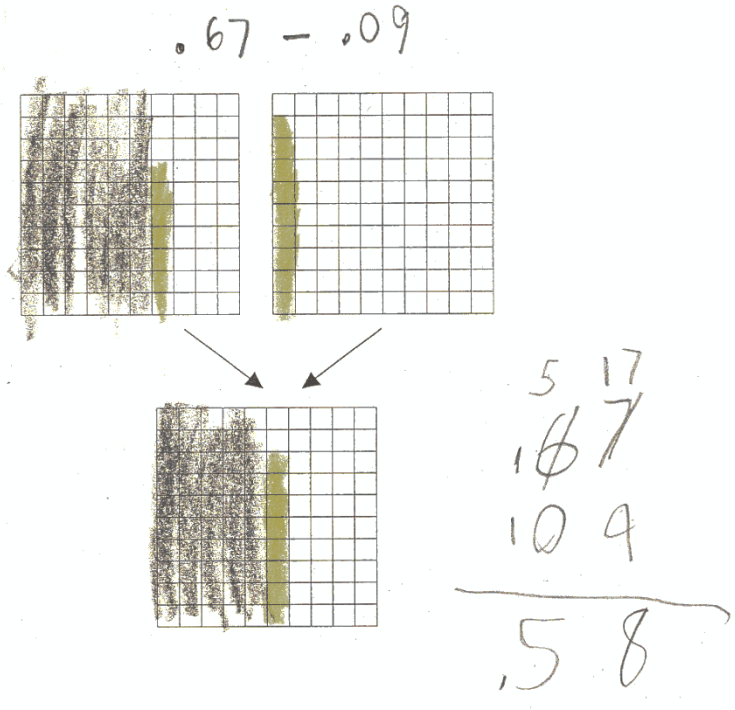
.67 - .09

To solve using the board the student first showed .67 on the left grid as 6 orange strips and 7 yellow squares. He showed .09 as 9 yellow squares on the right grid.

"For this one you have to subtract 7 to 9 which I am going to just use a bar and you would add a little circle there and get rid of one of the bars so it would be 5 bars and then go 1, 2, ,,,8 and there you go."

He then shaded in 5 oranges and 8 yellow on the bottom grid. "That would be 58 hundredths".

The student reflected on the two representations for the numbers involved in the problem. He imagined taking away 1 tenth and leaving 1 yellow square. (He pointed to the grid on the left while explaining this).

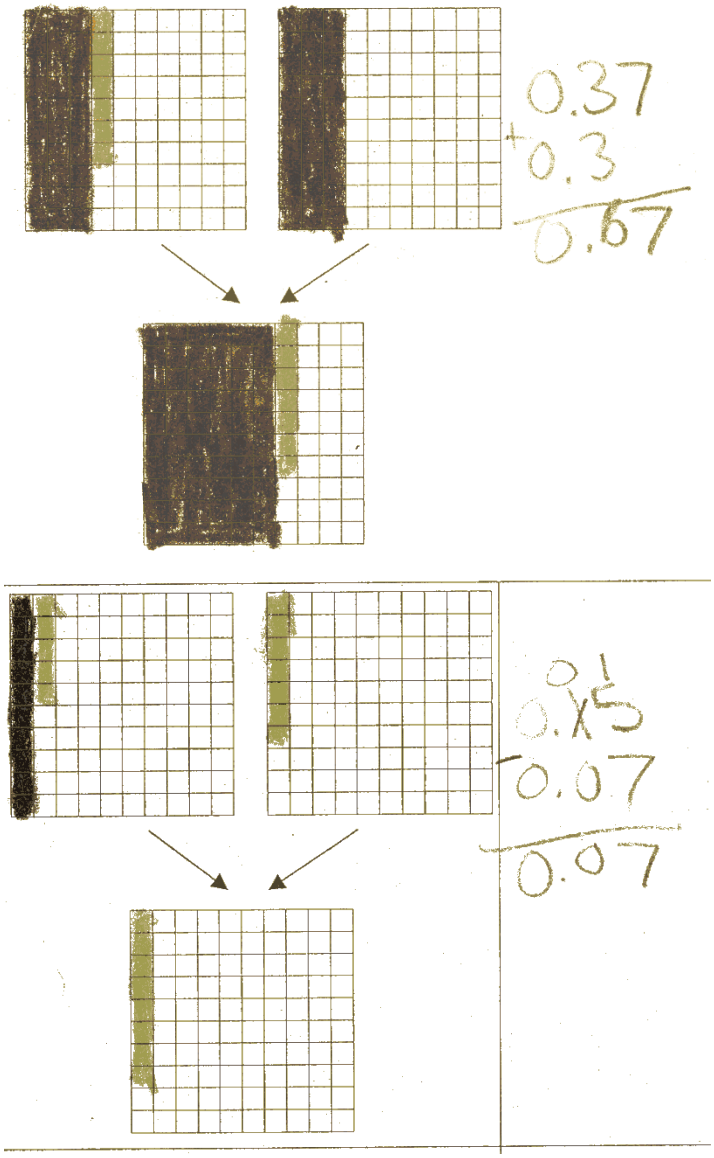


Students used the grids in different ways to solve these problems. The most important part of the grid is that it allows students to clearly represent the two numbers in the problem. Then the action on the two numbers can be done mentally or physically by writing on the boards. The bottom grid is used then to show the change after the action.

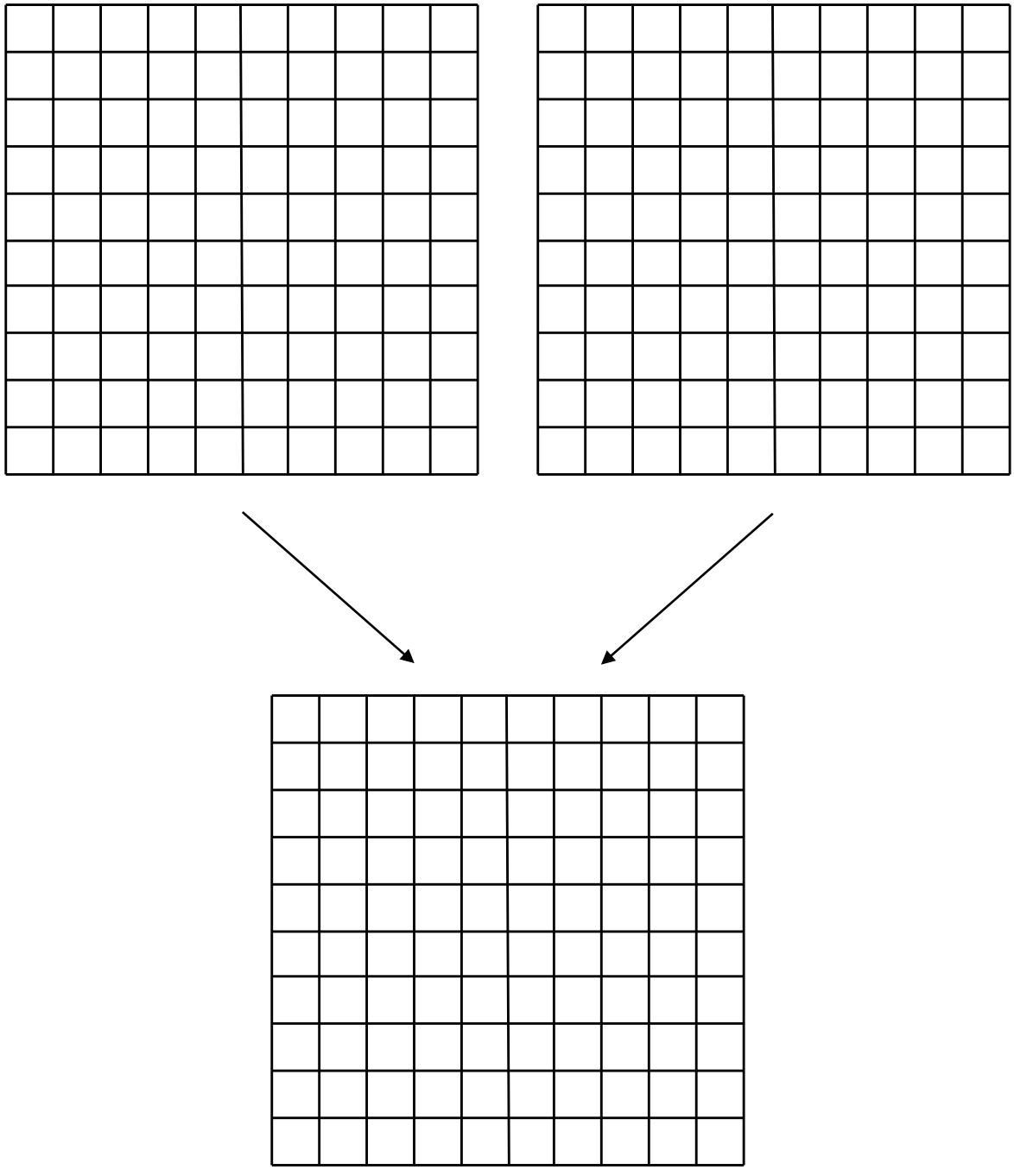
We found that students were able to accurately record the problems using symbols in vertical format after only a few examples. When doing the class work for lesson 11, many students first showed the two numbers on the top part of the addition board and then accurately recorded the problem in vertical form. Students did not make whole number errors of misaligning the numbers even though we did not introduce the rule "line up the decimal points".
Once the problem was written in symbols, many students solved the problem symbolically and then recorded the answer using orange and yellow colors on the bottom grid. The translations: symbolic to picture to symbolic aren't always that straight forward. Often we found that students use symbolic understanding first and then connected back to pictorial or concrete

representation. The reality is that multiple representations for the mathematical concepts/skills being developed enhance students understanding and the connections among different representations are varied.

Below find examples of student's work from lesson 11. There are no indications on the addition/subtraction board that the student used them to act out the problem. But the problems are recorded correctly using symbols and correctly solved.



Decimal Addition and Subtraction Board
Student Page A



37-hundredths plus 4-tenths	Greater than $\frac{1}{2}$ Less than $\frac{1}{2}$ Greater than 1 Less than 1	
56-hundredths take away 9-thousandths	Greater than $\frac{1}{2}$ Less than $\frac{1}{2}$	
1 and 6-hundredths plus 7-tenths	Greater than 1 Less than 1 Greater than 2 Less than 2	
1 and 58-hundredths take away 6-tenths	Greater than 1 Less than 1 Greater than $\frac{1}{2}$ Less than $\frac{1}{2}$	
73-hundredths take away 6-hundredths	Greater than 1 Less than 1 Greater than $\frac{1}{2}$ Less than $\frac{1}{2}$	

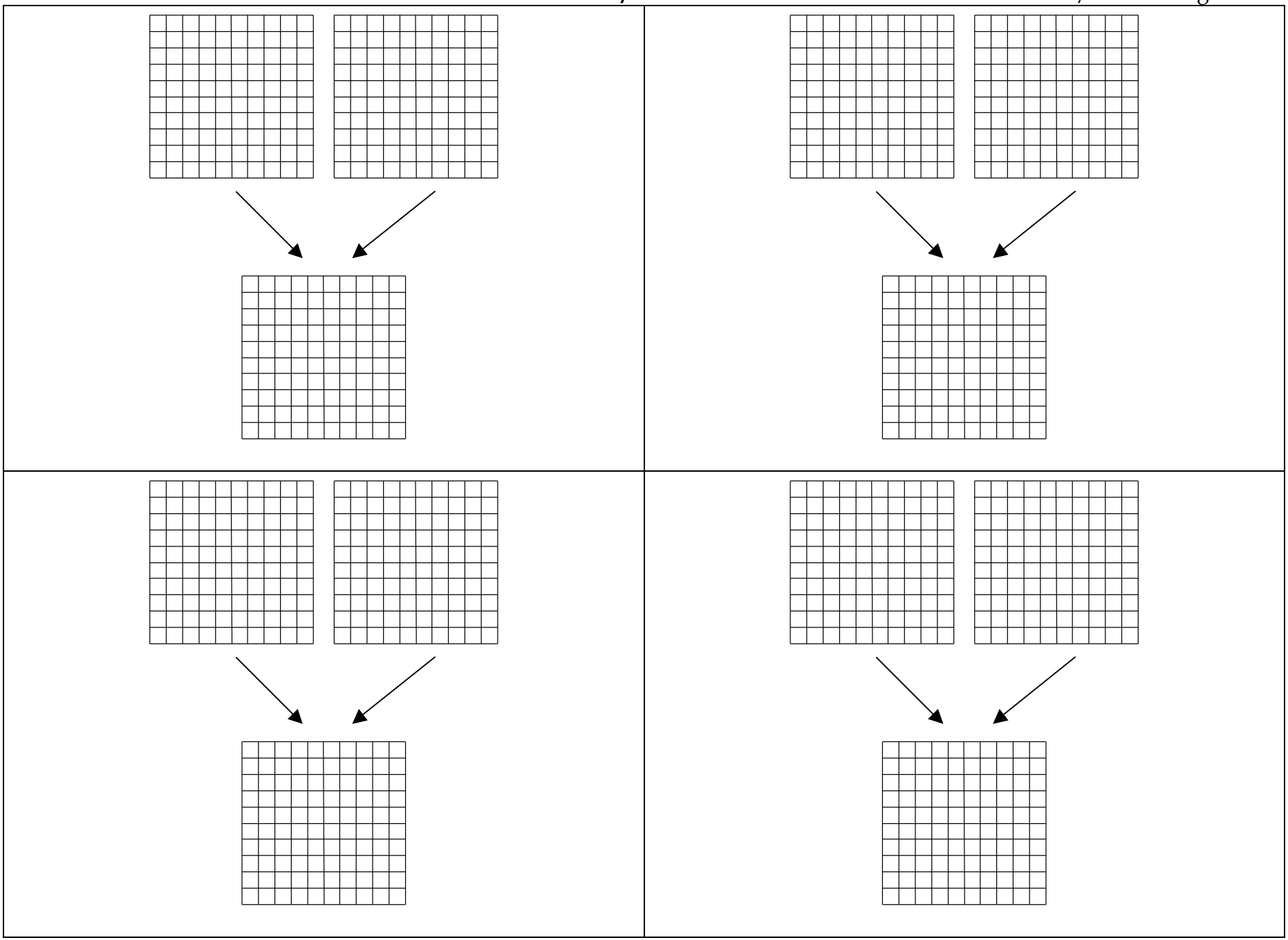
Imagine adding $\frac{1}{2} + \frac{1}{4}$ using fraction circles. Describe how you would solve this.

Imagine $\frac{1}{2}$ as a decimal, how much of the 10 x 10 grid is shaded?

Describe this amount if you used only orange.

Imagine $\frac{1}{4}$ as a decimal, how much of the 10 x 10 grid is shaded?

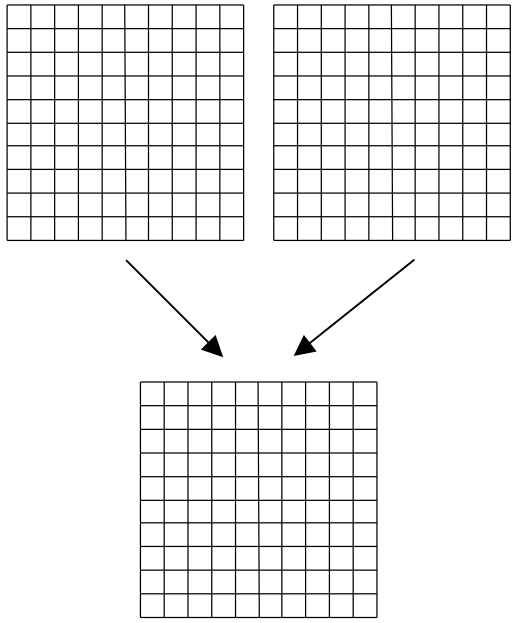
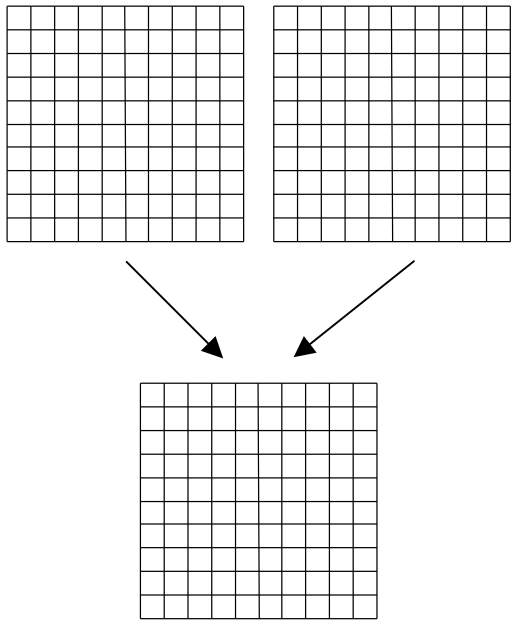
How many orange bars and yellow squares would be shaded to show $\frac{1}{4}$?



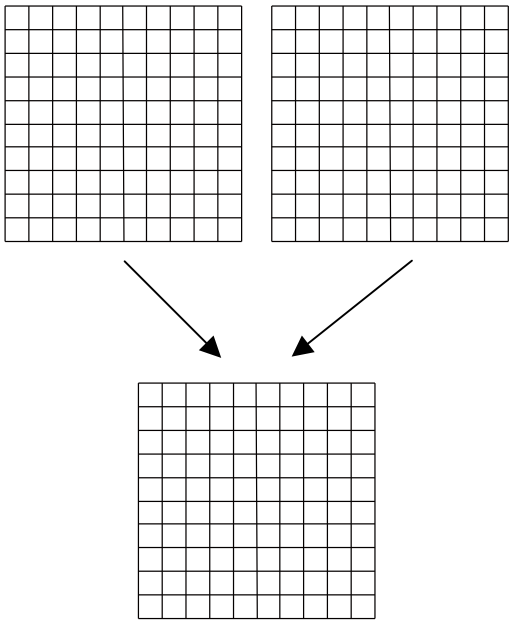
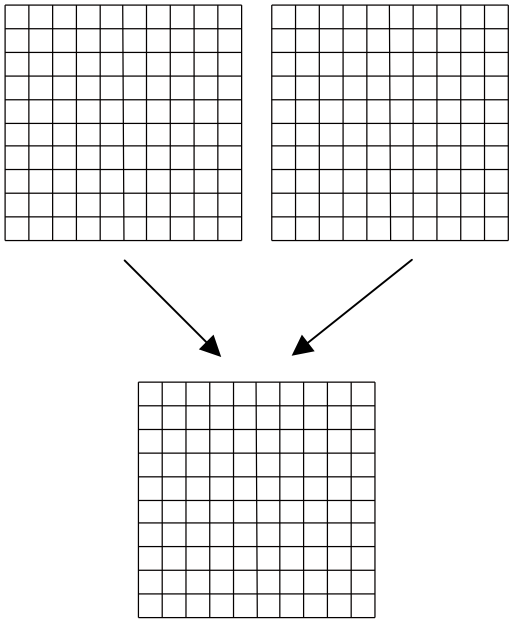
Name_____ Lesson 11 / Student Page B

Addition and Subtraction with 10 x 10 Grids

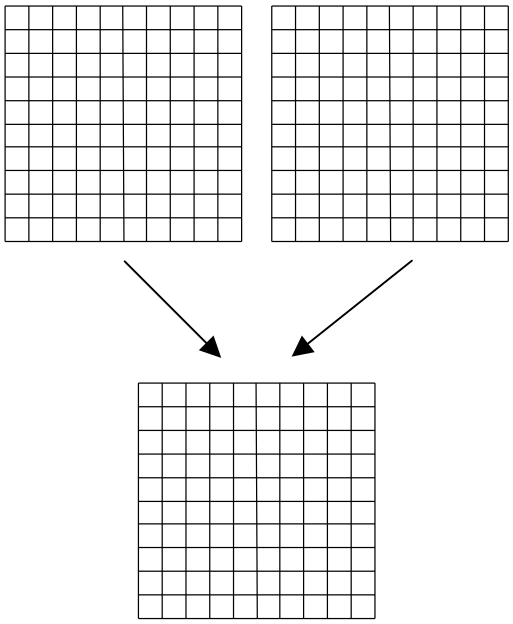
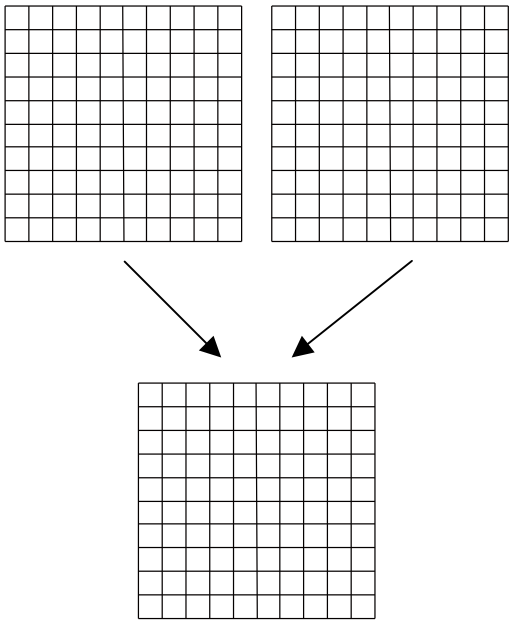
Solve each problem using the grid. Record the problem with symbols showing the final answer.

Problem	Estimate	Picture	Symbols
0.75 + 0.02 =	Greater than one or less than one?		
37-hundredths and 3-tenths more	Greater than $\frac{1}{2}$ or less than $\frac{1}{2}$? Greater than one or less than one?		

Name_____ Lesson 11 / Student Page C

15-hundredths minus 7-hundredths	Greater than $\frac{1}{2}$ or less than $\frac{1}{2}$? Greater than $\frac{1}{4}$ or less than $\frac{1}{4}$?		
0.75 – 0.4 =	Greater than $\frac{1}{2}$ or less than $\frac{1}{2}$?		

Name_____ Lesson 11 / Student Page D

99-hundredths – 88-hundredths	Greater than $\frac{1}{2}$ or less than $\frac{1}{2}$? Greater than $\frac{1}{4}$ or less than $\frac{1}{4}$?		
57-hundredths minus 3-tenths	Greater than $\frac{1}{2}$ or less than $\frac{1}{2}$? Greater than $\frac{1}{4}$ or less than $\frac{1}{4}$?		

Post Lesson Reflection

Lesson_____

1) Number of class periods allocated to this lesson: _____

2) Student Pages used: _____

3) Adaptations made to lesson: (For example: added extra examples, eliminated certain problems, changed fractions used)

4) Adaptations made on Student Pages:

5) To improve the lesson I suggest: